

Project Name: IoT Smart Energy Meter
Document Status: **Project Charter & Overview**

Project: IoT Smart Energy Meter

Project Charter & Overview

Version: Final

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1 PROJECT BACKGROUND

Modern energy management demands precise, real-time data beyond what traditional power meters provide. To achieve this, this project develops an advanced Smart Energy Meter prototype designed specifically for Single-Phase AC and DC applications. Unlike most of the commercial units connected directly to mains power, this device is engineered as a household safety-focused low-voltage system operating strictly below 40V. By utilizing an isolation transformer to simulate mains conditions, the project allows for the simulated testing of signal processing and measurement accuracy in a safe, controlled environment.

Architected around a PIC24 microcontroller, the system moves beyond prefabricated modules by custom-designed voltage and current sensing circuitry. These inputs enable the precise computation of Real Power, RMS values, and energy consumption. For telemetry, the device prioritizes reliability by using an Ethernet interface to upload real-time data to a dashboard. This approach eliminates wireless instability and establishes a robust proof-of-concept for professional-grade power monitoring hardware.

2 PROJECT PURPOSE / OBJECTIVES

The primary goal of this project is to design, fabricate, and test a **low-voltage (<40V) prototype** of a Smart Energy Meter. This device will serve as a safer, scalable educational model for industrial power monitoring, utilizing a PIC24 microcontroller and custom signal conditioning.

SMART Objectives:

- **Measurement:** Design custom sensing circuitry to measure Single-Phase AC and DC loads, strictly limited to < **40V** (via isolation transformer) for safety.
- **Accuracy:** Compute Real Power (P), RMS Voltage V_{rms} , and RMS Current I_{rms} with **±2% accuracy** to validate the custom sensor design.
- **Connectivity:** Establish a hardwired **Ethernet** connection to transmit real-time telemetry to a custom dashboard, ensuring data stability superior to wireless alternatives.
- **Interface:** Develop a responsive **touchscreen UI** on the device to allow users to adjust the screen brightness and time & date also display real-time polyphase measurements, system status and has active led status for power, data logging, system down/failure, ethernet connectivity.
- **Safety:** Implement hardware and firmware protection to detect over-voltage and over-current events (> 40V, >10A) and trigger immediate alerts to the dashboard.
- **Completion:** Complete all fabrication, coding, and final demonstration of the functional prototype by **April 10, 2026**, within a **\$200 budget**.

Measurability & Timeframe:

- **Validation:** Accuracy verified against lab equipment (DMM, Oscilloscope) using resistive and inductive loads within the 40V and 10A limit.
- **Stability:** System proven to log and upload data via Ethernet continuously for at least **24-hour burn-in period** without data loss.
- **Deadlines:** Full hardware assembly and firmware integration completed for the final April 2026 deadline.

This project aims to engineer a safety-focused, Ethernet-connected energy meter prototype. By limiting the input voltage to **< 40V**, the project focuses on high-fidelity signal processing and reliable data transmission without the hazards of mains voltage. The device uses a PIC24 microcontroller to process data from custom sensors, providing proof-of-concept for professional-grade energy monitoring hardware.

3 MEASURABLE OUTCOMES

- **Verified Accuracy:** Achieved measurement accuracy within $\pm 2\%$ for Real Power (P), RMS Voltage V_{rms} , and RMS Current I_{rms} under safe low-voltage operation (**< 40V**).
- **Successful Demonstration:** Hardware proven capable of measuring:
 - Single-Phase AC loads (moderated via isolation transformer to **< 40V AC**).
 - DC loads (limited to **< 40V DC**).
- **User Interface:** Functional touchscreen UI displays real-time AC/DC measurements, system status, and active alarm indicators.
- **Reliability & Telemetry:** Continuous data logging and telemetry transmission via **Ethernet** for a minimum of 24 hours without data loss or connection drops.
- **Safety Compliance:** Validation of hardware protection circuits ensuring input voltage is clamped or isolated to never exceed the **40V safety limit**, with automated alerting if thresholds are approached.
- **Documentation:** Complete engineering documentation package including PIC24 firmware source code, custom sensor schematics (including transformer/protection stage), PCB layout, Bill of Materials (BOM), and validation test reports.

4 KEY STAKEHOLDERS' SUMMARY

Role / Entity	Responsibilities & Communications Plan
Project Sponsor (College Dept / Capstone Committee)	* Responsibilities: Provides final approval on the project scope and budget. Grades the final deliverable. * Communication: Monthly progress reports and final presentation in April 2026.
Project Manager	* Responsibilities: Maintains the schedule (Gantt chart), manages the budget, and ensures the team sticks to the scope. Responsible for ordering parts from suppliers.

<i>(Student Leader)</i>	* Communication: Lead weekly team stand-ups and serve as the main point of contact for the Advisor.
Faculty Advisor <i>(Professor/Instructor)</i>	* Responsibilities: Provides technical guidance on high-voltage safety. Reviews Altium schematics before manufacturing to prevent costly errors. * Communication: Bi-weekly design reviews and office hour consultations.
Component Suppliers <i>(DigiKey)</i>	* Responsibilities: Supply specific electronic components (e.g., PIC24 MCU, Ethernet interface hardware, precision analog parts) and manufacture the custom PCBs within the 2-week lead time. * Communication: Transactional (Order confirmations, shipping tracking).
Lab Technician <i>(Nathaniel)</i>	* Responsibilities: Maintain the soldering stations and 3D printers required for assembly. May assist with complex reflow soldering if needed. * Communication: requests for equipment access or safety supervision.

5 PROJECT MILESTONES

Analysis and Planning Complete: Jan 19, 2026 *(Completion of Research, BOM Selection, and Initial Circuit Design)*

Breadboard Prototyping Complete: Jan 29, 2026 *(Completion of Analog Voltage/Current sandboxes and Safety tests)*

Hardware Design Complete: Mar 02, 2026 *(Completion of PCB Layout, Manufacturing during Reading Week, and physical Assembly)*

Firmware and Software Complete: Feb 27, 2026 *(Completion of Measurement Core, Ethernet Stack, and Touchscreen UI drivers)*

Assembly, Integration & Testing Complete: Mar 17, 2026 *(Completion of "Smoke Tests," Calibration to $\pm 2\%$, and Reliability Stress Testing)*

Final Demo & Documentation Sign Off: Apr 08, 2026 *(Submission of Final Report and Live Demonstration to Committee)*

6 MAJOR DELIVERABLES

- **Project Planning Package:** Charter, Overview, Block Diagram, WBS, Gantt Chart, Design Considerations.
- **Hardware Design Package:** Schematics, PCB layout, Bill of Materials (BOM), and fabrication files for the custom PIC24-based mainboard.
- **Firmware Package:** C source code, build scripts, and configuration files for the PIC24 MCU.
- **Backend / Dashboard:** Configuration for the hardwired Ethernet telemetry and local/remote dashboard interface.
- **Testing & Calibration Report:** Test cases, validation results against lab equipment, and calibration data for custom sensors.
- **Final Report, Presentation, Flyer Poster.**

Shipping Times: The project schedule depends on DigiKey shipping our specific analog components and microcontroller, and how fast the factory can manufacture and ship our custom circuit boards (estimated 1 to 2 weeks).

Lab Availability: We depend on the college engineering labs being open for students to use the soldering stations and 3D printers.

At the end of the project, the client can expect the following:

A Working Prototype: One fully assembled, safety-limited (<40V) device inside a 3D-printed case. It connects to a low-voltage isolated source (step-down transformer), powers its logic via DC input, and measures AC/DC power using custom discrete circuitry.

Design Files: Complete circuit diagrams and board layout files created in Altium Designer, featuring the custom voltage and current sensing stages.

Source Code: All C programming code used to run the PIC24 MCU, including Input Capture logic for Power Factor and Ethernet stack integration.

Final Report & Presentation: A document explaining the "discrete component" design approach, sensor calibration, safety limitations (<40V), and instructions on using the dashboard.

7 BUDGET

The project operates on a strict student budget, with an estimated projected total cost of **\$192.00**. To manage expenses, the team will prioritize high-quality signal acquisition hardware (PCB and Sensors) while utilizing college resources for fabrication tools.

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Category	Item Description	Estimated Cost (CAD)
PCB Fabrication	Custom 4-Layer PCB (Required for Ethernet signal integrity & noise reduction in analog lines).	\$60.00
Microcontroller	PIC24FJ1024GB610 (16-bit MCU) + Crystal Oscillators + Programming Header.	\$12.00
Analog Sensing	Precision Shunt Resistors (0.1Ω, 1%), Rail-to-Rail Op-Amps and high-tolerance voltage divider resistors.	\$15.00
Connectivity	Ethernet Controller + MagJack (RJ45 connector with built-in magnetics).	\$12.00
HMI Display	Touchscreen Module (Same as previous power supply project) for real-time dashboard visualization.	\$30.00
Safety Power	DC/DC Converter + Input Protection (Fuse holders, TVS Diodes, etc).	\$28.00
Enclosure	PLA/PETG Filament for 3D printing the custom safety case (pro-rated cost of material).	\$10.00
Contingency	Backup Funds (15%) for component damage, replacement parts, or shipping overages.	\$25.00
TOTAL		\$192.00

8 ASSUMPTIONS

- **Lab Equipment:** We assume the college will provide the soldering stations, oscilloscopes, and function generators (critical for validating custom analog circuits), as well as the 3D printers needed to fabricate the enclosure.
- **Relevant Software Licenses:** We assume a licensed version of **Altium Designer** is available through the college computers for PCB layout.
- **Component Availability:** We assume that the specific **PIC24 microcontroller**, **precision Op-Amps** and **Ethernet interface components** will remain in stock at major electronics distributors (DigiKey).
- **Teamwork:** We assume each member will dedicate at least 8 hours/week in the lab and 8 hours/week outside of college to ensure the project meets the April deadline.

9 RISK ASSESSMENT MATRIX

Probability	Impact		
	Low	Medium	High
High	R1	R6	R3
Medium		R5	R2
Low			R4

Risk Descriptions:

R1: Component Sourcing & Lead Times (High Probability / Low Impact)

Description: Sourcing specific discrete analog components ex. 0.1% tolerance resistors, specific Rail-to-Rail Op-Amps or the specific PIC24 MCU may encounter stock shortages at distributors.

Mitigation:

- Finalize the BOM immediately (week 3 or earlier).
- Select pin-compatible alternatives for Op-Amps and Regulators during the schematic phase.
- Order "backup" quantities (e.g., 5x required amount) for small passives.

R2: PCB Fabrication & Signal Integrity (Medium Probability / High Impact)

Description: Unlike using pre-made modules, we are designing the sensor layout ourselves. Risks include incorrect footprints (preventing assembly) or Ethernet signal interference (rendering the network interface non-functional).

Mitigation:

- Use 4-Layer PCB stack-up to provide dedicated Ground/Power planes, shielding sensitive analog signals from digital noise.
- Perform a "Paper Doll" test (printing the PCB layout 1:1) to verify footprints before ordering.
- Utilize Altium Designer DRC (Design Rule Check) features strictly.

R3: Measurement Inaccuracy / Analog Noise (High Probability / High Impact)

- Description: Custom shunt-based circuits are highly susceptible to noise, ground loops, and thermal drift, which may cause the meter to fail the $\pm 2\%$ accuracy objective.
- Mitigation:
 - Implement software averaging RMS calculation over 64+ samples to filter noise.
 - Include calibration potentiometers or software offset constants.

R4: Input Protection Failure / Safety (Low Probability / High Impact)

- Description: Although limited to 24V AC, accidental connection of a higher voltage source or failure of the isolation transformer could damage the low-voltage PIC24 microcontroller.
- Mitigation:
 - Install TVS Diodes and Fuses directly at input terminals to clamp spikes $>40V$.
 - Use an isolation transformer for all AC testing; never connect the prototype directly to a wall outlet.

R5: Firmware Complexity - Ethernet Stack (Medium Probability / Medium Impact)

- Description: Integrating a raw Ethernet stack is significantly more complex than using Wi-Fi modules, risking project delays if the team struggles with the protocol handling.
- Mitigation:
 - Use existing Microchip Code Configurator (MCC) libraries rather than writing drivers from scratch.
 - Fallback Plan: If Ethernet fails, implement UART/Serial logging to a PC as an acceptable "Backup Deliverable."

R6: Knowledge & Skill Gap (High Probability / Medium Impact)

- Description: The shift from "plug-and-play" modules to discrete analog design (Op-Amps, filters) and PIC24 architecture exceeds the team's prior experience with simpler Arduino-based projects. There is a risk of stalling due to a lack of theoretical understanding of differential sensing or 16-bit register manipulation.
- Mitigation:
 - Faculty Consultation: Schedule bi-weekly reviews specifically for the analog front-end design with an electronics professor.
 - Modular Prototyping: Breadboard the Analog Sensing Stage individually and verify it with an oscilloscope before designing the full PCB.
 - Reference Designs: Strictly adhere to proven architecture rather than attempting novel circuit topologies.

10 CONSTRAINTS

Time Limit: The project must be finished and ready to demonstrate by April 10, 2026.

Safety Rules: The device operates strictly under 40V yet we must enforce strict PCB clearance and to prevent over-current faults and ensure professional laboratory safety.

Manual Assembly: We must be able to assemble the board by hand. We cannot use parts that are too small for human hands to solder.

Programming Language: All firmware must be written in C, which requires careful memory management compared to easier languages like Python.

Limited lab and fabrication access: schedules, shared equipment.

Time: 13 weeks of term, with reading break and holidays; limited lab access windows

11 CHALLENGES

Discrete Analog Signal Integrity: Achieving $\pm 2\%$ accuracy using custom Op-Amp and Shunt Resistor circuits is significantly more difficult than using pre-made modules. Challenges include mitigating noise, managing ground loops on the custom PCB, and calibrating for component tolerances.

Mixed-Signal PCB Design: Designing a 4-layer PCB that combines sensitive analog sensor signals with high-speed digital Ethernet noise requires careful layout (stack-up, star grounding) to prevent interference.

Firmware Concurrency (Real-Time vs. Network): The PIC24 must handle high-speed continuous sampling (for RMS/Power Factor math) while simultaneously maintaining a stable TCP/IP (Ethernet) connection and driving the Touchscreen UI & LED program without bugs & misc. issues.

Ethernet Stack Integration: Implementing the wired Ethernet interface can be more complex than standard Wi-Fi libraries in some ways, posing a risk of integration delays.

Safe Prototype Constraints: Strictly limiting the input to $<40V$ requires robust hardware protection (TVS diodes, fuses) to guarantee the device remains safe even if accidental over-voltage events occur during testing.

Component Sourcing & Schedule: Sourcing specific high-precision discrete components (0.1% resistors, rail-to-rail Op-Amps) and awaiting custom 4-layer PCB fabrication leaves little margin for error in the project timeline.

12 COMMUNICATION PLAN

Weekly status updates to instructors (Logbook and oral report).

Team meetings: At least $2 \times$ per week (1 planning, 1 technical).

Issue tracking: Simple shared documents or tracker listing tasks, owners, and blockers.

Design reviews: At least one schematic review and one layout review with peers/instructor before ordering PCBs.

13 PROJECT CHARTER ACCEPTANCE

Sponsor / Instructor: _____ Date: _____

Project Manager: _____ Date: _____

Team Member(s): _____ Date: _____

PROJECT OVERVIEW

1. Background

This project focuses on engineering a high-fidelity Smart Energy Meter prototype designed to measure Single-Phase AC and DC loads. Unlike standard consumer meters that rely on prefabricated sensor modules, this device emphasizes fundamental engineering design by utilizing a PIC24 microcontroller and custom discrete analog circuitry. We will design a custom 4-layer circuit board using Altium Designer and write the firmware in C to handle real-time signal processing and Ethernet communication.

Instead of connecting directly to high-voltage mains, the device operates as a safety-focused laboratory prototype, strictly limited to inputs under 40V. It utilizes precision shunt resistors and voltage dividers to measure power metrics with high accuracy. Rather than hosting a simple local webpage, the system is designed for industrial reliability, transmitting real-time telemetry via a hardwired Ethernet interface to a comprehensive dashboard for remote monitoring and analysis.

2. Purpose and Scope

Functional Objectives:

- **Implement Measurement Modes:**
 - Single-Phase AC (50/60 Hz) via isolation transformer.
 - DC circuits (e.g., batteries, lab power supplies).
 - Support measurement update periods of 1 second.
- **Compute and Display:**
 - RMS Voltage V_{rms} and Current I_{rms} .
 - Real (P), Apparent (S), and Reactive (Q) power.
 - Power Factor and Frequency.
 - Energy Direction
 - Import and Export energy
 - Accumulated Energy counters (kWh).

- **System Status & Feedback:**
 - **Interactive Touchscreen HMI:** time, date and Alert Status.
 - **Hardware Status Indicators:** Implement a 4-color LED system for immediate system feedback:
 - **Green:** System Power On/Off
 - **Red:** System Failure / Fault Detected.
 - **Yellow:** Data Logging Activity (Write in Progress).
 - **Blue:** Ethernet Link Established.
- **Support Features:**
 - **Power Management:** Implement PIC24 "Idle" and "Sleep" modes to reduce power consumption during inactivity.
 - **Ethernet Telemetry:** Transmit real-time data packets to a remote dashboard.
 - **Protection Logic:** Automated hardware control output (Relay) to disconnect load if Voltage > 40V or Current > 10 A.

Ambition:

- Engineer a discrete-component laboratory prototype that demonstrates fundamental analog signal processing skills.
- Move beyond pre-made modules by designing custom Op-Amp and Shunt-based sensing stages.
- Serve as an educational tool to reinforce fundamental electrical measurement concepts and demonstrate Ethernet stack implementation on the PIC24 architecture.

Engineering & Documentation:

- Provide complete 4-layer PCB layout files, custom analog schematics, and Bill of Materials (BOM).
- Deliver fully documented PIC24 C source code, including Input Capture drivers, LED state logic, and Ethernet integration.
- Provide a detailed Calibration Report validating the accuracy of the custom analog front-end against lab equipment.

3. Requirements

Functional Requirements:

- The system shall measure single-phase AC voltage and current with accuracy better than $\pm 1\%$ voltage and $\pm 2\%$ current, within the specified <40V range.
- The system shall measure DC voltage and current with similar accuracy.
- The system shall compute P, Q, S, PF, and frequency for each phase every 200ms.
- The system shall accumulate energy consumption (Wh, kWh) and retain values across power cycles.
- The system shall log measurements to internal flash memory at 1-minute intervals.
- The system shall indicate operational status via physical LEDs (Green=ON, Red=Fail, Yellow=Log, Blue=Net).

- The microcontroller shall enter a Low-Power Idle state when not actively sampling or operating of touchscreen display.

Non-Functional Requirements:

- **Safety:** Strict adherence to <40V input limit; on-board protection via Fuses and TVS Diodes.
- **Reliability:** Capable of continuous 24-hour operation for lab demonstrations without system hang-ups (enforced via Watchdog Timer).
- **Maintainability:** Firmware shall be modular, separating the Measurement Core, UI Logic, and Ethernet Stack.
- **Usability:** Clear, responsive UI with simple plug-and-play Ethernet configuration (DHCP).
- **Cost:** BOM cost target between \$150–\$200.

4. Target Audience

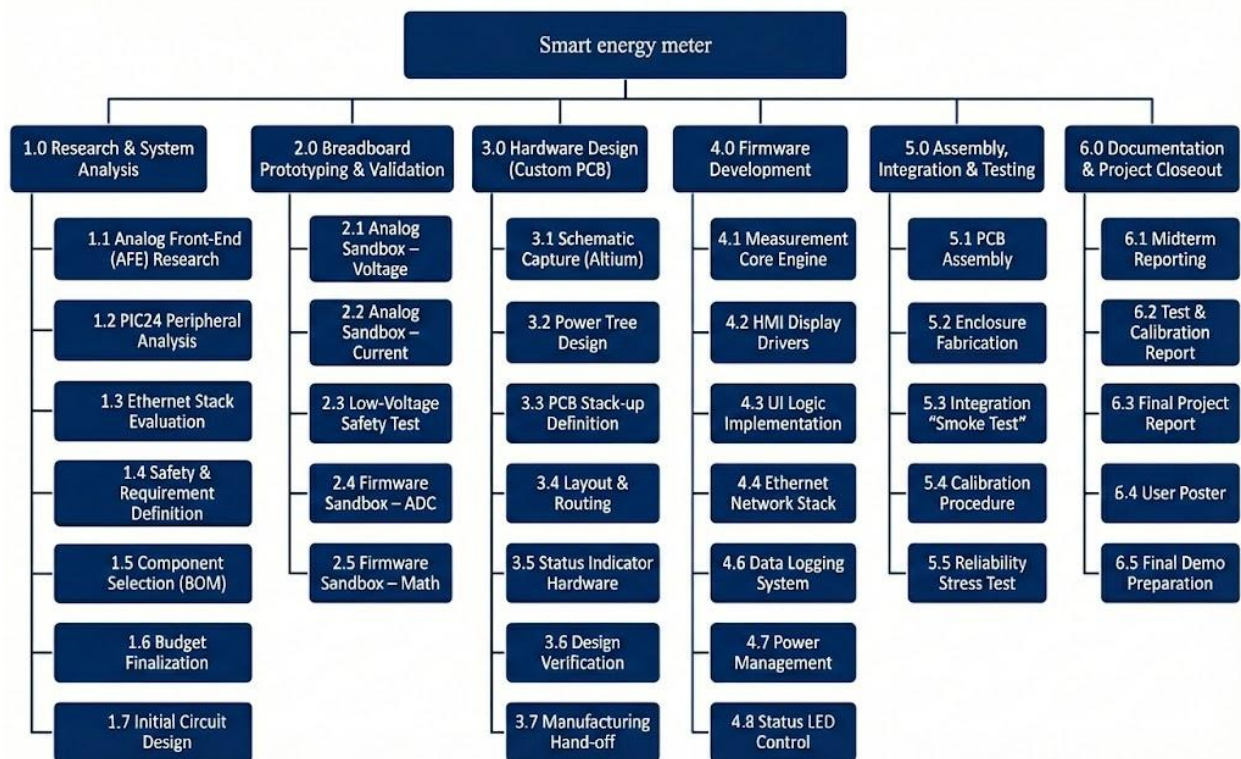
- **Primary:** Electronics Engineering (ELEN) students and instructors needing a transparent, component-level platform for teaching power measurement, discrete analog design, and embedded systems.
- **Secondary:** Educational institutions requiring safe, low-voltage lab equipment to demonstrate AC power theory without high-voltage safety risks.

5. Expected Outcomes

At the end of the project, the expected deliverables and outcomes are:

- A fully functional Discrete Analog Power Meter prototype meeting all safety and accuracy requirements.
- Demonstrated Measurement of Single-phase AC and DC loads with $\pm 2\%$ accuracy.
- Operational Feedback via a responsive Touchscreen HMI and the coded 4-color LED Status system.
- Remote Connectivity established via Ethernet, showing live data on a remote dashboard.
- Comprehensive Design Package including the Altium PCB files, C Source Code, and Calibration Test Report.

WORK BREAKDOWN STRUCTURE



1.0 Research & System Analysis

- **1.1 Analog Front-End (AFE) Research:** Research discrete Op-Amp topologies for differential voltage sensing and calculate required Shunt Resistor wattage (I^2R) for thermal stability.
- **1.2 PIC24 Peripheral Analysis:** Review the PIC24 datasheet to confirm pin mapping for the **Input Capture (IC)** module (required for Power Factor timing) and SPI availability for the Touchscreen.
- **1.3 Ethernet Stack Evaluation:** Evaluate the Microchip Code Configurator (MCC) TCP/IP Lite stack versus custom UDP drivers to determine the most stable approach for telemetry.
- **1.4 Safety & Requirement Definition:** Define exact voltage (<40V) and current trip thresholds for the protection logic and select appropriate TVS diode clamping voltages.
- **1.5 Component Selection (BOM):** Finalize the Bill of Materials, selecting precision 1% resistors, Rail-to-Rail Op-Amps, and verifying stock status at DigiKey/Mouser.
- **1.6 Budget Finalization:** Update the cost sheet with real-time pricing for the 4-layer PCB fabrication and specific discrete components.
- **1.7 Initial Circuit Design:** Draft the preliminary schematics for the Voltage Divider, Current Shunt Amplifier, and Power Regulation stages to guide the breadboard prototyping phase.

2.0 Breadboard Prototyping & Validation

- **2.1 Analog Sandbox – Voltage:** Build the resistive voltage divider and Op-Amp buffer circuit on a breadboard; validate signal linearity with an oscilloscope.
- **2.2 Analog Sandbox – Current:** Build the Shunt Resistor and Non-Inverting Amplifier stage; test for noise levels and ground loop issues.
- **2.3 Low-Voltage Safety Test:** Test the external isolation transformer under load to ensure stable 24V AC output and verify fuse/diode protection triggers correctly.
- **2.4 Firmware Sandbox – ADC:** Write test firmware to configure the PIC24 ADC and read raw values from the breadboard circuit via UART logging.
- **2.5 Firmware Sandbox – Math:** Prototype the RMS calculation algorithm in C using the raw breadboard data to verify math accuracy.

3.0 Hardware Design (Custom PCB)

- **3.1 Schematic Capture:** Create the full circuit diagram in Altium Designer, including the LED Status Logic (current-limiting resistors and transistor drivers) for the 4-color indicator system.
- **3.2 Power Tree Design:** Design the on-board power regulation stage with proper filtering capacitors.
- **3.3 PCB Stack-up Definition:** Configure a 4-Layer stack-up to shield sensitive analog traces from Ethernet noise.
- **3.4 Layout & Routing:** Route impedance-controlled differential pairs for the Ethernet interface and place Op-Amps close to sensor inputs.
- **3.5 Status Indicator Hardware:** Select and place surface-mount LEDs for specific system feedback:
 - Green: System Power on/off.
 - Red: System Failure/Fault.
 - Yellow: Data Logging Activity.
 - Blue: Ethernet Link/Activity.
- **3.6 Design Verification:** Run Electrical Rule Checks, Design Rule Checks, and perform a "Paper Doll" 1:1 print test to verify component footprints.
- **3.7 Manufacturing Hand-off:** Generate Gerber X2 files, Drill files, and Pick-and-Place data; submit order to fabrication house.

4.0 Firmware Development

- **4.1 Measurement Core Engine:** Develop the interrupt-driven ADC sampling routine (circular buffers) and the main loop for RMS/Power/Energy calculations.
- **4.2 HMI Display Drivers:** Write code for the Touchscreen and create the graphics library for drawing gauges and text.
- **4.3 UI Logic Implementation:** Implement the state machine for switching between "Dashboard View," "Trend View," and "Settings/Alerts" screens.

- **4.4 Ethernet Network Stack:** implement the UDP/TCP socket logic for transmitting telemetry to the dashboard.
- **4.5 Protection Logic:** Code the real-time safety checks to trigger the Relay output immediately if Voltage > 40V or Current > 10A.
- **4.6 Data Logging System:** Implement the file system or raw storage logic to save minute-by-minute logs to the internal flash memory.
- **4.7 Power Management:** Implement Power Saving Modes (Idle/Sleep) for the PIC24 microcontroller to reduce consumption when measurement or display is inactive.
- **4.8 Status LED Control:** Program PIC24 to manage the physical status indicators:
 - Code toggle on & off logic for Green (Power ON).
 - Code latching logic for Red (Fault).
 - Code blink pattern for Yellow (Data Logging).
 - Code link-detection logic for Blue (Ethernet).

1.0 Assembly, Integration & Testing

- **5.1 PCB Assembly:** Solder the power regulation and PIC24 MCU sections first; verify 3.3V rails and In-Circuit Serial Programming connection. Solder the precision Op-Amps, Shunts, Ethernet Jack, and Connectors.
- **5.2 Enclosure Fabrication:** Design & 3D print the protective case and mount the PCB, Transformer, and Touchscreen securely.
- **5.3 Integration "Smoke Test":** Power up the fully assembled unit, flash the full firmware image, and verify boot sequence and screen response.
- **5.4 Calibration Procedure:** Apply known reference loads to the device, compare against a Lab Multimeter, and derive software offset values.
- **5.5 Reliability Stress Test:** Run the device continuously for 24 hours to check for memory leaks, thermal drift, or network connection drops.

6.0 Documentation & Project Closeout

- **6.1 Midterm Reporting:** Compile prototyping data and initial design files into the Midterm Status Report and Presentation.
- **6.2 Test & Calibration Report:** Document the final accuracy results, oscilloscope captures of noise performance, and calibration constants.
- **6.3 Final Project Report:** Author the comprehensive report detailing the Discrete Analog design choices, Firmware architecture, and Safety validation.
- **6.4 User Poster:** Create a poster guide explaining how to set up the device and how the device works, configure Ethernet settings, and interpret the Dashboard.
- **6.5 Final Demo Preparation:** Script and rehearse the live demonstration for the Capstone Committee.

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ID	Task Name	Dependencies	Optimistic (O)	Regular (R)	Pessimistic (P)	Expected (te)
1	Research & System Analysis					
1.1	Analog Front-End Research	<i>Start</i>	1	2	3	2
1.2	PIC24 Peripheral Analysis	1.1	1	2	3	2
1.3	Ethernet Stack Evaluation	1.2	1	3	5	3
1.4	Safety & Requirement Definition	1.1	1	1	2	1.2
1.5	Component Selection (BOM)	1.4	1	2	3	2
1.6	Budget Finalization	1.5	0.5	1	1.5	1
1.7	Initial Circuit Design	1.5	2	3	5	3.2
2	Breadboard Prototyping					
2.1	Analog Sandbox – Voltage	1.7	2	4	6	4
2.2	Analog Sandbox – Current	1.7	2	4	6	4
2.3	Low-Voltage Safety Test	1.4	1	1	2	1.2
2.4	Firmware Sandbox – ADC	1.2, 2.1	2	4	6	4
2.5	Firmware Sandbox – Math	2.4	3	5	8	5.2
3	Hardware Design (Custom PCB)					
3.1	Schematic Capture (Altium)	2.1, 2.2	3	5	8	5.2

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3.2	Power Tree Design	3.1	1	2	3	2
3.3	PCB Stack-up Definition	3.1	0.5	1	1.5	1
3.4	Layout & Routing (4-Layer)	3.2, 3.3	5	8	12	8.2
3.5	Design Verification (Paper Doll)	3.4	1	2	3	2
3.6	Mfg Hand-off (Express Ship)	3.5	6	9	12	9
4	Firmware Development					
4.1	Measurement Core Engine	2.5	8	12	18	12.3
4.2	HMI Display Drivers	3.1	3	6	9	6
4.3	UI Logic Implementation	4.2	4	7	10	7
4.4	Ethernet Network Stack	1.3	5	10	15	10
4.5	Protection Logic	2.3	1	3	5	3
4.6	Data Logging System	4.1	2	4	6	4
5	Assembly, Integration & Testing					
5.1	PCB Assembly (Soldering)	3.6	2	3	5	3.2
5.2	Enclosure Fabrication	3.4	2	4	6	4
5.3	Integration "Smoke Test"	5.1	1	1	2	1.2
5.4	Calibration Procedure	5.3	3	5	7	5
5.5	Reliability Stress Test	5.4	1	2	3	2
6	Documentation & Closeout					

6.1	Midterm Reporting	2.5	2	4	6	4
6.2	Test & Calibration Report	5.5	2	3	4	3
6.3	Final Project Report	6.2	4	7	10	7
6.4	User Poster	4.3	2	3	4	3
6.5	Final Demo Preparation	6.3	2	4	6	4

Pert Time estimate = $(o + 4m + p)/6$

o = optimistic time, m = regular estimate time, p = pessimistic time

- **Schedule Feasibility Analysis**

Objective:

To verify that the project Critical Path fits within the academic calendar (January 5, 2026 – April 10, 2026) after accounting for all statutory holidays and university closures.

- **Total Available Time Budget**

First, we determine the hard constraint on the maximum number of working days available.

- **Project Start Date:** January 5, 2026
- **Project End Date:** April 10, 2026
- **Total Duration:** 14 weeks (approximately 70 weekdays)

Less Statutory Holidays & Closures:

- Reading Week (Feb 16 – Feb 20): –5.0 days
- Easter Break (Apr 3 & Apr 6): –2.0 days

$$T_{\text{available}} = 63.0 \text{ working days}$$

- **Required Execution Time (Critical Path Calculation)**

The Critical Path (T_{required}) is defined as the sum of the expected durations (t_e) for the longest chain of dependent tasks. If these tasks are delayed, the project fails to meet the deadline.

$$T_{\text{required}} = \text{Analysis} + \text{Prototyping} + \text{Design} + \text{Fabrication} + \text{Assembly} + \text{Testing} + \text{Reporting}$$

- **Analysis Phase (Tasks 1.1–1.7):** 7.2 days
- **Prototype Phase (Parallel Tasks 2.1 & 2.2):** 4.0 days
- **Schematic Design (Task 3.1):** 5.2 days

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Document Status: **Project Charter & Overview**

- **PCB Layout (Task 3.4):** 8.2 days
- **Manufacturing & Shipping (Task 3.6):** 9.0 days
- **PCB Assembly (Task 5.1):** 3.2 days
- **System Integration (Task 5.3):** 1.2 days
- **Calibration & Metrology (Task 5.4):** 5.0 days
- **Final Reporting (Task 6.3):** 7.0 days
- **Demo Preparation (Task 6.5):** 4.0 days

$$T_{\text{required}} = 54.0 \text{ working days}$$

- **Variance and Conclusion**

The project buffer is calculated to assess resilience against unforeseen delays.

Buffer Calculation:

$$T_{\text{available}} - T_{\text{required}} = 63.0 - 54.0 = \mathbf{9.0} \text{ working days}$$

Safety Margin:

$$(9.0/54.0) \times 100 \approx \mathbf{16.7\%}$$

Conclusion:

The project is feasible. The calculated Critical Path of 54.0 days fits within the 63.0-day academic window, leaving a 9.0-day (approximately two-week) safety buffer. This feasibility relies on executing the Analog Voltage and Current prototyping tasks in parallel and utilizing expedited shipping for PCB manufacturing during the February Reading Week.